

Business Practice Manual for Outage Management

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Revision History

Version	Date	Description
2	1/1/2010	PRR 87 Revisions due to Standard Capacity Product. 7.3.1 Resource Adequacy Resources with Pmax Between 1.0 MW and 10 MW
3	8/10/2010	PRR 156 Revision due to Proxy Demand Resource 7.3.2 Outage reporting for Proxy Demand resources.
4	6/15/2011	PRR 428 Revision to include Seven Day Requirement for Planned Transmission Outage Requests.
5	9/26/2011	PRR 453 Revision to include 7.3.3 Reliability Demand Response Resource (RDRR) outages and derates shall be managed and reported to SLIC in the same manner as generator resources. Tariff effective April 1, 2012.
6	3/13/2012	PRR 522 Revision to include: 4.2.1.2 Sharing Outage Information with Natural Gas Suppliers



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Version	Date	Description
		PRR 524 Revision to include: 7.3.4 Non Resource Specific RA resources Outage Reporting

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[Appendix A. NDA and Use Information Agreement for Sharing of Generation and Electric Transmission Outage Information With A Natural Gas Utility](#)

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1. Introduction

Welcome to CAISO *BPM for Outage Management*.

In the *Introduction* section you will find the following information:

- The purpose of CAISO BPMs
- What you can expect from this CAISO BPM
- Other CAISO BPMs or documents that provide related or additional information

Please note that the business processes described in this BPM are also covered in several CAISO Operating Procedures in much more detail. If there is an apparent conflict between this BPM and the CAISO Operating Procedures listed in Section 1.5, the Operating Procedures take preference.

1.1 Purpose of California ISO Business Practice Manuals

The Business Practice Manuals (BPMs) developed by CAISO are intended to contain implementation detail, consistent with and supported by the CAISO Tariff, including: instructions, rules, procedures, examples, and guidelines for the administration, operation, planning, and accounting requirements of CAISO and the markets. Exhibit 1-1 lists CAISO BPMs.

Exhibit 1-1: CAISO BPMs

Title
BPM for Market Operations
BPM for Market Instruments
BPM for Settlements & Billing
BPM for Scheduling Coordinator Certification & Termination
BPM for Congestion Revenue Rights
BPM for Candidate CRR Holder Registration
BPM for Full Network Model

Title
BPM for Rules of Conduct Administration
BPM for Outage Management
BPM for Metering
BPM for Reliability Requirements
BPM for Credit Management
BPM for Compliance Monitoring
BPM for Definitions & Acronyms
BPM for BPM Change Management
BPM for the Transmission Planning Process

1.2 Purpose of this Business Practice Manual

The *BPM for Outage Management* covers the business processes used by CAISO to coordinate the scheduling of generation and transmission maintenance outages within the jurisdiction established in the CAISO Tariff.

The *BPM for Outage Management* includes the following:

- A description of the objective of the Outage Management business processes, the roles of the CAISO and Facility Owners of Generating Units and sections of the CAISO Controlled Grid
- A description of the CAISO's Outage Coordination Office and Control Center and their respective roles in the Outage Management business processes
- A summary of the CAISO long-range planning process used to develop an integrated rolling 12-month generation and transmission plan of maintenance Outages
- A summary the CAISO analysis of proposed Generating Unit Outages to determine aggregate generation availability and overall adequacy
- A description of the processes and timing used by Participating Generators and Participating Transmission Owners (TOs) to submit new or revised Outage requests and for CAISO to process these requests and to approve or reject them



➤ A description of the requirements for scheduling and approving Regulatory Must Take Generator Outages

- A description of the need for and process used to request and receive Final Approval before initiating an Approved Outage
- A discussion of the CAISO management of Forced Outages and the process for requesting and receiving an extension to an Outage
- A summary of the communication requirements, including the need for a single point of contact, use of the SLIC application and a backup mechanisms if SLIC is unavailable and the use of voice communication with the CAISO Control Center for certain processes
- A description of the process for inspection of CAISO Outage records and the notice requirements for gaining access to those records
- A review of the CAISO generating adequacy reports that are developed on an annual, quarterly and monthly basis
- A description of the requirement for reporting Generating Unit Forced Outage Reports within a 48-hour period
- A description of the CAISO Questionable Outage Report to FERC and a summary of the CAISO criteria
- A description of the SLIC application used to support most Outage Management business processes
- A description of the processes used to reflect Outage data in the Day-Ahead Market under normal and exception conditions

The provisions of this BPM are intended to be consistent with the CAISO Tariff. If the provisions of this BPM nevertheless conflict with the CAISO Tariff, the CAISO is bound to operate in accordance with the CAISO Tariff. Any provision of the CAISO Tariff that may have been summarized or repeated in this BPM is only to aid understanding. Even though every effort will



be made by the CAISO to update the information contained in this BPM and to notify Market Participants of changes, it is the responsibility of each Market Participant to ensure that he or

she is using the most recent version of this BPM and to comply with all applicable provisions of the CAISO Tariff.

A reference in this BPM to the CAISO Tariff, a given agreement, any other BPM or instrument, is intended to refer to the CAISO Tariff, that agreement, BPM or instrument as modified, amended, supplemented or restated.

The captions and headings in this BPM are intended solely to facilitate reference and not to have any bearing on the meaning of any of the terms and conditions of this BPM.

1.2 References

The definition of acronyms and words beginning with capitalized letters in this BPM, are given in the *BPM for Definitions & Acronyms*.

Other reference information related to this BPM includes:

- CAISO Operating Procedures
- Generating Station Requirements and Communication (G-200)
<http://www.caiso.com/thegrid/operations/opsdoc/gcp/index.html>
- Schedule and Forced Outage Operating Procedure, No. T113
<http://www.caiso.com/thegrid/operations/opsdoc/transmon/index.html>
- *BPM for Market Operations*
(Link to be inserted in the future)
- Emergency Procedures
<http://www.caiso.com/thegrid/operations/opsdoc/emergency/index.html>



- SLIC Application Instructions

<http://www.caiso.com/docs/2002/01/29/2002012913530024587.pdf>

- Annual, Quarterly, and Monthly System Adequacy Reports

(Link to be inserted in the future)

2. Objectives, Roles, Scope, & Participants

In the *Objectives, Roles, Scope, & Participants* section you will find the following information:

- Objectives of the CAISO Outage Management related business processes
- Roles of the participants in those processes
- Scope of the Outage Management related business processes
- Participants in the Outage Management Processes
- CAISO entities involved in those processes

Tariff Sections 9.1, Coordination and Approval for Outages, 9.2, Responsibility for Authorized Work on Facilities, and 9.3 Coordination of Outages and Maintenance

2.1 Objective

The objective of the CAISO business processes related to Outage Management is to enable CAISO to coordinate transmission and generation Maintenance Outages as far as possible in advance to allow CAISO to:

- Maintain System Reliability
- Maximize Schedule feasibility relative to Congestion and meeting Demand
- Reflect Outage information in the CAISO Markets as soon as possible to allow Market Participants to effectively plan for the operation of their resources

2.2 CAISO Role

CAISO's role in the Outage Management business processes, as outlined in the CAISO Tariff, is to coordinate Outage schedules for maintenance, repair, and construction of Generating Units, sections of CAISO Controlled Grid, and Interconnections. This BPM describes the processes CAISO uses to perform this Outage scheduling coordination role.

2.3 Facility Owner Role

The Facility Owners of the Generating Units, sections of CAISO Controlled Grid and Interconnections that are subject to processes covered in this BPM, remain solely and directly

responsible for the performance of all maintenance work, whether on energized or de-energized facilities, including all activities related to providing a safe working environment.

In addition, the Facility Owners are responsible to schedule facility Outages and report Outage information consistent with the business practices and requirements included in the CAISO Tariff, in this BPM and specified in more detail in the CAISO Operating Procedures (See Section 1.5, References).

The Facility Owner may elect to have an agent perform some or all of the activities required to meet their responsibilities related to Outage Management, however, the Facility Owner remains responsible for the successful completion of these activities. See Section 6, Communication of Outage Maintenance Information, of this BPM for a discussion of the requirement for a Facility Owner to establish a single point of contact.

2.4 Scope

The *BPM for Outage Management* describes the business processes that apply to the Outage Planning of CAISO Controlled Generation, Grid and Interconnection System Equipment.

These include the following:

- Long Range Outage Planning
 - Long Range Annual Plan due October 15.
 - Quarterly Updates to the Long Range Plan are due each quarter.
 - January 15, April 15, July 15

Status of Long Range Outage Requests can be expected to be available by the 30th day after annual and or quarterly plans are due.

- All outage requests with a start date of 46 days or greater from date of submittal are process by the Long Range Outage Coordinator.
- Short Range Outage Planning
 - All outage requests with a start date of less than 46 days from date of submittal are process by the Short Range Outage Coordinator.
- Analysis and reporting of Generator maintenance plans
- Steps in Outage Coordination include:
 - Request submitted by SLIC web, email or phone.
 - Accept or enter request in SLIC.
 - Search SLIC for conflicts.
 - Search procedures and previous SLIC tickets for any limitations or requirements.
 - Update SLIC ticket with any requirements. Verify appropriate attributes are flagged.
 - Make the request pending if additional information is required (include what information is required in the SLIC ticket).
 - Send the pending request to the party providing the information with a request to respond back by a certain date.
 - If the request is pending, once the requested information is received update the request as needed and notify affected parties, revise the "Notice of Planned Intertie Limitation" notice and update the TTC sheet as required and any revised generation requirements and approve the SLIC ticket.
 - If there are no conflicts, approve the request.
 - Notify affected parties of the request (including as appropriate a "Notice of Planned Intertie Limitation"), Emailing the SLIC ticket internally to ISO and

update the TTC sheet if required. Notify affected parties of any generation that may be limited or required to be off line per Market rules.

- All Generation Outage Requests are reviewed for RA capacity commitment.
- All Transmission System Equipment Outages that could significantly affect CRR's must be submitted 30 days prior to the month the outage is to begin.
NOTE: "Significant System Equipment" has been defined by the ISO through a stakeholder process and has been added to section 4.2 of this BPM.



- All Participating Transmission Operator's and Participating Generators shall report equipment outages to the ISO in accordance with CAISO Tariff section 9.
- Management of Forced Outages
 - Outages that are submitted with less than 3 business days prior to the outage start day will be designated as short notice Forced Outages.
 - All Generator Forced Outages will be reviewed per CAISO Tariff section 9.3.10.5.
- Outage records and reporting
- Outage records are maintained in CAISO SLIC
 - Each outage is maintained in the CAISO SLIC Data Base using a unique I.D. number.
 - Each outage may have multiple requests attached to it and all requests are maintained in the CAISO SLIC Data Base using a unique I.D. number.

2.5 Application to Parties

The *BPM for Outage Management* applies to CAISO and the following parties:

- All Generators with a Participating Generating Agreement
- All Participating Transmission Owners



- Scheduling Coordinators/ Participating Generators
 - Request Outages via SLIC Web, SLIC/API and email if SLIC is unavailable.
 - Immediate Forced Outages are auto accepted by SLIC and reviewed by the real-time dispatchers.
- Connected Entities, to the extent the agreement between the Connected Entity and CAISO so provides
- Participating Transmission Operators
 - Request Outages via SLIC Web, SLIC/API and email if SLIC is unavailable.
 -
 - Immediate Forced Outages are automatically processed by SLIC and reviewed by the real-time dispatchers.

2.6 CAISO Outage Coordination Office

As outlined in the CAISO Tariff, the CAISO Outage Coordination Office (OCO) coordinates and processes all Maintenance Outage requests as described in this BPM. The CAISO OCO operates Monday through Friday, except holidays, and accepts, schedules, and approves or denies Maintenance Outage requests, as necessary, for the reliable operation of CAISO Controlled Grid.

The OCO personnel are located in Folsom, California. The location, contact information, and areas of responsibility for this office are detailed in the most recent version of the applicable CAISO Operating Procedures (See Section 1.5, References), which is available through the CAISO Website.

The OCO uses an electronic application to support the receipt, review, processing, and approval or rejection of requests for Outages. The electronic application used by the CAISO for Outage Management, referred to as SLIC (for **S**cheduling and **L**ogging for **I**SO of **C**alifornia), is briefly described in Section 6.2.4 of this BPM.

The type of scheduled work and Outages that the OCO coordinates and approves is shown in the following list. (This list is not an exhaustive list of work that requires CAISO approval.)

- Balancing Authority Area Interconnections work
 - All outages that affect interconnected systems will be coordinated between Interconnected Transmission Operators.
- All work on facilities forming the CAISO Controlled Grid, including associated control or protective equipment
 - All outages affecting CAISO PTO equipment and Generators with a participating generator agreement.
- All reportable Outages or partial curtailments of Participating Generators with a rated capacity greater than 1MW
- All reportable Outages of Reliability Must Run Generating Units
-
- EMS work that disables any portion of the CAISO Grid monitoring, control or protective equipment including EMS equipment and communication circuits
- EMS work that affects Generator AGC or RIG equipment or communication circuits
- Ancillary Service certification testing and compliance testing
- Interconnections with responsible entities outside the CAISO Balancing Authority Area

2.7 CAISO Control Center

The CAISO Control Center is responsible for the Real-Time aspects of Outage Management as described in detail in the CAISO Operating Procedure (See Section 1.5, References) and briefly discussed below. In general, the CAISO Control Center handles:

- Outage Management requests/events for Outages that are planned for the following day after the OCO office hands off the responsibility at the close of the OCO business day prior to the day of the outage (at 1500 Hours)
- Outage Management requests/events that relate to Approved Maintenance Outages in progress.



- Forced Immediate Outages received via SLIC are automatically approved and listed on an auto accept workspace in SLIC to be reviewed by the real-time dispatcher.

For example the CAISO Control Center performs the following functions:

- Reviews requests for Final Approval to initiate an Outage and provides Final Approval after appropriate review of the Outage is completed as described in Section 4.11 of this BPM
- Reviews requests for extensions to Approved Maintenance Outages
- Manages immediate Forced Outages of Generating Units

The CAISO Control Center reviews and processes other Outage Management related requests as described in this BPM and discussed in more detail in the CAISO Operating Procedure (See Section 1.5, References). The CAISO Control Center personnel use a combination of the SLIC application and phone communication to perform their Outage Management functions.

3. Long Range Planning of Maintenance Outages

In the *Long Range Planning of Maintenance Outages* section you will find the following information:

An overview of the process used for developing the CAISO Long Range Outage Plan of Generator and CAISO Controlled Grid and Interconnection Outages

3.1 Long Range Planning Process Overview

Each quarter, CAISO looks ahead to develop an integrated generation and transmission Maintenance Outage plan that incorporates the most up to date scheduling and load forecasting data. This integrated process uses load forecasts, Outage schedule requests submitted by Participating Generators and Participating TOs, previously Approved Maintenance Outages and other outage information to determine which of the requested Outages can be performed while maintaining adequate System Reliability over a rolling 12-month planning horizon. This process looks at generation and transmission Outages simultaneously to make certain that any generation Outages that affect transmission ratings or capabilities are considered when requests for transmission Outages are analyzed and visa versa.

3.2 Outage Requests & Updates

The following sections describe the processes used by Participating Generators and Participating TOs to submit requests and updates to previous requests for Outages to be included in the CAISO long-range planning process and by CAISO to process these requests and to develop a Long Range Outage Plan.

All information submitted to CAISO related to Outages must be submitted in accordance with Section 6, Communication of Maintenance Outage Information, of this BPM.

3.2.1 Long Range Outage Requests

By October 15th of each year, each Participating Generator provides the CAISO with any proposed Outages for the following calendar year for its Generating Units and each Participating TO provides CAISO with any proposed Outages for the following year for its transmission system. The proposed Outage submittals also include any requested additions to or changes to previously approved Outages. As a result, the submittal looks forward approximately 15 months, which includes any new or revised Outages for the period January 1st until December 31st of the



following year. The mechanism and processes used to submit proposed outages and for CAISO to process the proposed outages are described in detail in Section 4 of this BPM.

The following information is required for each Generating Unit Outage Request:

- Generating Unit name and Location Code
- MW capacity remaining available
- Scheduled start and finish time/date for the maintenance Outage
- Nature of the proposed maintenance Outage
- Note the required system outage boundaries to facilitate the equipment outage.
- For Outages where there is a possibility of some flexibility in when the Outage can be scheduled, submit the preferred start and end dates in the required fields and in the outage text box state the earliest start date and the latest finish date, along with the required duration of the outage

The following information is required for each transmission related Outage Request:

- Identification of the facility and location
- Nature of the proposed Maintenance Outage
- Scheduled start and finish time/date for the Maintenance Outage
- For Outages where there is a possibility of some flexibility in when the Outage can be scheduled, submit the preferred start and end dates in the required fields and in the outage text box state the earliest start date and the latest finish date, along with the required duration of the outage.
- All outages submitted into the long-range outage plan.
 - Select the "LT" attribute in the SLIC outage request
 - Outage Plans submitted from the 1st of the month due to the 15th due date shall be considered as equal opportunity and be processed as described in CAISO operating procedure T-113

- Outages submitted between Long Range and Quarterly Updates shall be considered as “First Come First Serve” priority.
- If a previously approved outage is rescheduled by the equipment operator and the new schedule falls outside of the original requested date the original request priority will be lost and the reschedule will be considered as a new outage request.
- Outage Management and Operations Engineering will review all Long Range Outage Requests for reliability assessment.
- Status of outage requests shall be available to requester by 30 working days after the Long Range Outage Plan due date.
- Outage requests that are disapproved shall have a short reason for the disapproval and a suggested date when the outage could be approved for.
- Approved Outages that require final outage details added within the month prior to the start date shall be highlighted, so they may be queried and completed by the Short Range Outage Coordinators.
- All outages are reviewed on a weekly basis for reliability and changing system conditions. Outages may be rescheduled or cancelled due to changing system conditions.
- Outages processed by the Long Range Outage Coordinator will be passed along to the Short Range Outage Coordinators when the outage start date is less than 46 days away.
- Each day outages automatically roll from the Long Range into the Short Range.

Short Range Outage Coordination will accept process and approve short notice outage requests and will further process outages rolled into the short range from long range. Approval of short range requests depends on the ability to fit the outages into the outage plan received from the long-range process and maintains system reliability.

3.2.2 Quarterly Update to Planned Outages

By the close of business on the 15th day of January, April, and July, each Participating Generator and Participating TO provides CAISO with quarterly updates of its proposed Outages for use by CAISO in its Long Range Outage Plan process. These updates must identify known changes to any previously planned Outages and any additional Outages anticipated over the



next 12 months from the time of the quarterly update. The data required for each of these updates is the same as discussed in Section 3.2.1 of this BPM. For example, the January 15th update includes any Outages for the period April 1st of the year of the submittal year to March 31st of the following year.

3.2.3 Additional Information Requests

CAISO may request additional information or seek clarification from any Participating Generator or Participating TO regarding the information submitted with a proposed Outage. This additional information may be used to assist CAISO in prioritizing conflicting requests for Outages.

3.2.4 Reporting For Regulatory Must-Take Generation

Information regarding planned Outages for resources providing Regulatory Must-Take Generation must be provided to CAISO OCO by the Participating TO or UDC having an Existing Contract with such resource or by a Participating Generator. Information provided is that obtained by the Participating TO, UDC or a Participating Generator pursuant to the terms of the Existing Agreement with the Regulatory Must-Take Generation resource or as requested by CAISO.

3.3 Balancing Authority Area Coordination

Tariff Section 9.3.10.6, Other Balancing Authority Areas

CAISO coordinates the exchange of proposed ISO Controlled Grid Maintenance Outages, as appropriate with the Operators of adjacent Balancing Authority Areas, and uses the procedures and processes for maintenance outage scheduling and posting that are developed with each adjacent Balancing Authority Area.

3.4 CAISO Review of Proposed Maintenance Outages

Under the CAISO long term planning process, the OCO reviews the Maintenance Outages submitted for the 15-month period to determine if any one or a combination of Maintenance Outage requests relating to CAISO Controlled Grid and Interconnection facilities, Generating Units or System Units may cause CAISO to violate the Applicable Reliability Criteria. This review takes into consideration factors including, but not limited to, the following:

- Forecast peak Demand conditions
- Other Maintenance Outages, previously Approved Maintenance Outages, and anticipated unit Outages
- Potential to cause Congestion
- Impacts on the transfer capability of system facilities or Interconnections
- Impacts on the Market

If CAISO determines that any of the proposed Maintenance Outages are forecast to cause CAISO to violate the Applicable Reliability Criteria, CAISO notifies the relevant requestor of the violation. The Participating Generator or Participating TO then must revise the proposed Maintenance Outage(s) and resubmit the Outage to CAISO pursuant to Section 4 of this BPM.

3.5 Approval of Outages

The information relating to each Maintenance Outage submitted by a Participating Generator or Participating TO in accordance with Section 4 of this BPM constitutes a request for a Maintenance Outage and is not considered an Approved Maintenance Outage until CAISO has notified that Participating Generator or Participating TO of such approval. Once a request for a planned Generator Outage has been received by CAISO, CAISO provides a response to the requester within a period of one month as described in Section 4. Also as described in Section 4, the Outage request can be approved or denied and, if denied, CAISO may suggest that the entity consider requesting the Outage be placed in the Pending State.

Such an Approved Maintenance Outage is part of the CAISO's Long Range Outage Plan and unless the Outage is changed the only step the Participating Generator or Participating TO must take prior to initiating the Outage is to request and receive Final Approval using the process described in Section 4.11 of this BPM.

3.6 CAISO Analysis of Generating Unit Proposed Outage Plans

Tariff Section 9.3.6.4B, CAISO Analysis of Generating Unit Outage Plans



As a part of the CAISO long-range planning process, CAISO considers all approved and requested Outages and other data as appropriate, to calculate the aggregate Generation capacity projected to be available in the following time frames:

- On an annual and quarterly basis, CAISO calculates the aggregate weekly peak Generation capacity projected to be available during each week of the following year and quarter, respectively
- On a monthly basis, CAISO calculates the aggregate daily peak Generation capacity projected to be available during the month

Reports that are based upon this analysis and that are published by CAISO are described in Section 7 of this BPM.

4. Requesting & Seeking Approval for Maintenance Outages

In the *Requesting & Seeking Approval for Maintenance Outages* section you will find the following information:

- A discussion of the requirements for Regulatory Must-Take Generation
- An overview of the processes used to propose new Maintenance Outages and to revise or cancel Approved Maintenance Outages
- A discussion of the processes used by the CAISO to review, approve or reject proposed Outages
- A description of the process used to request Final Approval for an Outage and for CAISO to process the request

Tariff Section 9.3, Coordination of Outages and Maintenance

4.1 Regulatory Must-Take Generation

Tariff Section 9.3.5.2A, Communication of Approval or Rejection

Scheduling and approval of Maintenance Outages for resources providing Regulatory Must-Take Generation are coordinated as detailed in the applicable contract with the Participating TO or UDC, provided the Regulatory Must-Take Generator has not executed a Participating Generator Agreement. The Participating TO or UDC must advise CAISO OCO of scheduled and approved Maintenance Outages on resources providing Regulatory Must-Take Generation pursuant to Existing Contracts. If the Regulatory Must-Take Generator executes a Participating Generator Agreement, it must comply with all sections of this BPM, which are applicable to Participating Generators including but not limited to Section 3, Long Range Planning of Maintenance Outages and Section 4, Requesting & Seeking Approval for Maintenance Outages, of this BPM.

4.2 Participating Generator & Participating Transmission Owner Requests

This section describes the processes used by Participating Generators and Participating TOs to submit requests for new Outages or revisions to Approved Maintenance Outages and the processes used by CAISO to process requests and to approve or reject these requests.

4.2.1 Scheduling Requirements

Transmission Outage Scheduling

Each Participating TO must submit a new Maintenance Outage or a revision to an Approved Maintenance Outage to CAISO for approval. Such an Outage request must be submitted to the CAISO OCO via the SLIC application no later than Seven Days prior to the start date of the proposed Outage for Transmission facilities, as specified in CAISO Tariff Section 9.3.6.3.1. Outages submitted no later than seven days prior to start date can expect to receive a response by three business days prior to the outage start date. Note: The determination of seven day prior notice excludes the date of submission and the date of the outage. Transmission Outages effecting CRR revenue adequacy must be submitted no later than 30 days prior to the calendar month of the Outage of a Transmission Outage as specified in CAISO Tariff Section 9.3.6.3.2. As previously described in Section 3, Long Range Planning of Maintenance Outages, Participating TOs are asked to submit their request well in advance, to allow them to be analyzed with other proposed outages in the CAISO's long-range planning process.

For Outages of the CAISO Controlled Grid or Interconnections, the request must specify the following:

- Identification and location of the transmission system element(s) to be maintained
- Nature of the maintenance to be performed
- Note the required system outage boundaries to facilitate the equipment outage.
- Date and time the Maintenance Outage is to begin
- Date and time the Maintenance Outage is to be completed
- Time required to terminate the maintenance and restore the transmission system to normal operation



- Primary and alternate telephone numbers for the Operator's single point of contact

In the case of a request for a change to an Approved Maintenance Outage, the date and time of the original Approved Maintenance Outage

- Generation Outage Scheduling

Each Participating Generator must submit a new Maintenance Outage or a revision to an Approved Maintenance Outage to CAISO for approval. Such an Outage request must be submitted to the CAISO OCO via the SLIC application no later than Three Business Days prior to the start date of the proposed Outage for all Generating facilities, as specified in CAISO Tariff Section 9.3.6.3.1. As previously described in Section 3, Long Range Planning of Maintenance Outages, Participating Generators are asked to submit their request well in advance, to allow them to be analyzed with other proposed outages in the CAISO's long-range planning process.

For Generators the request for an Outage must specify the following:

- Generating Unit or System Unit name and Location Code
- Nature of the maintenance to be performed
- Date and time the Outage is to begin
- Date and time the Outage is to be completed
- State Resource Adequacy commitment -- In the notes section of the SLIC Outage Request, indicate:
 - Is there an RA commitment
 - If so, capacity and dates of RA commitment

- Time required to terminate the Outage and restore the Generating Unit to normal capacity
- Primary and alternate telephone numbers for the Operator's single point of contact
- In the case of a request for a change to an Approved Maintenance Outage, the date and the time of the original Approved Maintenance Outage

4.2.1.1Planned Outage Scheduling of Transmission Equipment Significant to Congestion Revenue Rights

The CAISO proposes to define outages to "Significant Facilities" as having a significant effect upon CRR revenue adequacy when the outage impacts the normal energy transfer capability or flow between points on the CAISO Controlled Grid. "Significant Facilities" include only Transmission facilities of the CAISO Controlled Grid:

1. rated above 200 kV (except for CAISO-approved exceptions), or
2. that are part of any defined flow limit as described in a CAISO operating procedure (except for CAISO-approved exceptions), or
3. that were out of service in the last three years and for which the CAISO determined a special flow limit was needed for real-time operation.

A list of the transmission facilities less than 200KV and that satisfy criteria (2) and (3) above is provided in the Operating Procedures. The list will be initially created in collaboration with the respective PTOs and will be reviewed by the CAISO in collaboration with the PTOs on an annual basis or as needed (determined by the ISO) and revised as appropriate; provided however, that the CAISO will ultimately determine the equipment included in the list.

The CAISO will apply the "30-day rule" for scheduling planned outages of Significant Facilities. Under the 30-day rule, requests for planned outages of Significant Facilities must be submitted to CAISO Outage Coordination at least 30 days prior to the start of the calendar month for which the outage is planned to begin. If the 30th day falls on a non ISO business day then the Planned Outage Request is due on or before the last business day, 30 days prior to the month the outage is to begin. This rule is intended to provide good outage data far enough in advance to allow the CAISO to reflect them in the network model used for releasing CRRs and thereby minimize impacts to the revenue adequacy of CRRs.

The CAISO emphasizes that the 30-day rule is not intended to prevent needed maintenance on significant facilities in circumstances where the 30-day rule cannot be followed without

adversely affecting the grid reliability. Accordingly, the following exceptions to the 30-day rule will be used to approve outages without classifying them as forced:

1. Outages that are less than 24 hours in duration.
2. Outages previously approved by CAISO that are moved within the same calendar month either by the CAISO or by request of the PTO.
3. CAISO approved allowable transmission maintenance activities during restricted maintenance operations as covered in CAISO operating procedure E-509A.
4. The most current list of specific Significant Transmission Equipment will be covered in CAISO operating procedure T-113.

The CAISO will review the history of these types of exceptions annually to determine whether they are effective in promoting adequate information for CRR purposes, and whether use of these exceptions should continue to be classified as planned.

4.2.1.2 Sharing Outage Information with Natural Gas Suppliers.
The CAISO must coordinate outages and derates of natural gas supply equipment along with all CAISO Grid equipment to assure reliable system operations.
In accordance with CAISO Tariff section 20.4(c)(iv), the CAISO has the authority to provide confidential transmission and generator outage information to natural gas utilities, under suitable contractual limits and protection concerning the disclosure and use of confidential or proprietary information consistent with the form non-disclosure agreement attached as appendix [A] hereto.

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4.3 CAISO Outage Scheduling & Approval Process

In general, the following steps are used for any request by a Participating Generator or Participating TO to schedule a Maintenance Outage. This includes requests that are processed as a part of the CAISO long-range planning process as well as requests for an Outage that were not processed in the long-range planning process but still submitted in accordance with the timing specified in Section 4.2.1 of this BPM.

4.3.1 Generation Resource Start Up Time

Generation Maintenance Outages should not include start up time. Each generator's start up time is documented in the CASIO Master File and is considered to begin once the generator has been called on by the ISO or for a scheduled start up.



4.4 Confirmation of Receipt of Outage Request

CAISO OCO acknowledges receipt of each request for a new Outage or to revise an existing Approved Maintenance Outage using the SLIC application.

4.5 Review of Requested Outages & Priorities

CAISO OCO reviews Maintenance Outages submitted to determine if any one or a combination of Maintenance Outage requests relating to CAISO Controlled Grid and Interconnection facilities, Generating Units or System Units may cause CAISO to violate the Applicable

Reliability Criteria. This review takes into consideration factors including, but not limited to, the following:

- Forecast peak Demand conditions
- Other Maintenance Outages, previously Approved Maintenance Outages, and anticipated unit Outages
- Potential to cause Congestion
 - CRR consideration
- Impacts on the transfer capability of Interconnections
- Impacts on the market

If CAISO determines that any of the proposed Maintenance Outages would cause CAISO to violate the Applicable Reliability Criteria, CAISO cannot approve the requested Outage.

4.5.1 Outage Request Priorities

Prioritization of all Outage requests are on a first-come, first-served basis, with the exception of those requests that are received as a part of the CAISO long-range planning process, from the 1st until the 15th of the month, which CAISO deems to have been received at the same time.

4.6 Delay Due to Insufficient Data

CAISO OCO may delay or disapprove its approval of a requested Maintenance Outage if sufficient or complete information is not received by CAISO OCO within the time frames set forth in Section 4.2.1 of this BPM. In the event that insufficient or incomplete information is provided, the requestor is notified via SLIC.

4.7 Acceptance or Rejection of Outage Schedule

Based upon the CAISO OCO review of the requested Outage described earlier, CAISO then approves or rejects such request in accordance with the CAISO Operating Procedures (See Section 1.5, References). CAISO approval or rejection for each submitted Outage request is communicated by CAISO via the SLIC application.

For requests, which are submitted in a time frame to be considered as a part of the CAISO long-range planning process, CAISO provides a response to the request within a 30-day period following the long range submittal due date.

4.8 Pending/Scheduled State of Request

Outages that have been submitted to, and accepted by the CAISO for review, are ultimately either approved or rejected. After acceptance by the CAISO, and prior to CAISO Approval or Rejection, Outages are in the Pending State. Outages that are in the Pending State are not to be considered Approved. Only Outages that have been approved by the CAISO OCO or by CAISO Real-Time Operations can be considered Approved.

Upon notification from the CAISO OCO that an Outage may be rejected due to forecast system conditions, the respective Owner may request that the Outage remain in the Pending State until subsequent forecasts confirm the precluding system conditions, and thereby retain for the Owner the priority associated with the time/date of the original Outage request. Placing an Outage request into the Pending State for the purpose of retaining the priority of the original Outage request is at the sole discretion of the CAISO OCO. If the request to retain the Outage in the Pending State is denied the Outage request is rejected as described in Section 4.9.

4.9 Rejection Notice

In the event that the CAISO OCO rejects a proposed outage, it will issue a rejection notice that identifies CAISO's reliability, security and/or market concerns that are the basis for the rejection and suggests possible remedies or schedule revisions which might mitigate any such concerns.

4.10 Withdrawal or Modification of Request

The Participating Generator or Participating TO may withdraw an Outage request at any time prior to actual commencement of the Outage. The Participating Generator may modify a request at any time prior to receipt of any acceptance or rejection notice from CAISO OCO, but CAISO OCO has the right to reject such modified request for reasons of System Reliability, system security, complexity of the proposed modifications, or insufficient time to assess the impact of such modifications.

4.11 CAISO Final Approval

A Participating Generator or Participating TO must not initiate an Outage without receiving Final Approval of the Outage as described in this section.

On the day when an Approved Maintenance Outage is scheduled to commence, the Participating Generator or Participating TO must contact CAISO Control Center by phone or electronically using SLIC Web to request Final Approval of the Approved Maintenance Outage. The request must include the approved starting time and return time for the Outage. The process for securing such Final Approval is described in more detail in the CAISO Operating Procedure (See Section 1.5, References).

4.11.1 Withholding of CAISO Approval & Rescheduling of Outage

CAISO Control Center has the authority to withhold Final Approval for an Approved Maintenance Outage for reasons of System Reliability. In that event, CAISO Control Center will immediately notify the Operator of its intention to withhold Final Approval for the Approved Maintenance Outage. The Maintenance Outage that did not receive Final Approval must be rescheduled by the Participating Generator or Participating TO using the procedures outlined in Section 4 of this BPM.

4.12 Changes to Planned Maintenance Outages

A Participating Generator or Participating TO may cancel a previously approved planned Maintenance Outage or submit a request to change a previously approved planned Maintenance Outage at any time. Requests for such changes must include the information specified in Section 4.2.1 and be submitted in accordance with the timing requirements in Section 4.2.1 of this BPM.

However, if a Participating TO cancels an Approved Maintenance Outage after 5:00 a.m. of the day prior to the day upon which the Outage is scheduled to commence and CAISO determines that the change was not required to preserve System Reliability, CAISO may disregard the availability of the affected facilities in determining the availability of transmission capacity in the Day-Ahead Market. CAISO, however, notifies Market Participants and reflects the availability of transmission capacity as promptly as practicable.

4.13 Outages Cancelled at CAISO Direction

Tariff Sections 9.3.6.10, Cancellation of Approved Maintenance Outage and 9.3.7, Maintenance Outage Requests by the CAISO.

CAISO OCO may, by providing notice no later than 0500 hours of the day prior to the day upon which the Outage is scheduled to commence, direct the Operator to cancel an Approved

Maintenance Outage, when necessary to preserve or maintain System Reliability. If such cancellation occurs after the 0500 hours deadline, CAISO may be liable to compensate the applicable Participating TO or Participating Generator, pursuant to the provisions of CAISO Tariff Section 9.3.7.3 for the direct and verifiable costs incurred by that Participating TO or Participating Generator as a result CAISO's cancellation of an Approved Maintenance Outage. The Operator, acting in accordance with Good Utility Practice, must comply with CAISO's direction. CAISO will give notice of any such direction to Market Participants prior to the deadline for submission of Bids in the DAM for the day on which the Outage was to have commenced.

4.14 Outages at CAISO Direction

CAISO may at any time request that an unscheduled Maintenance Outage or change to an Approved Maintenance Outage be scheduled and taken by a Participating Generator or Participating TO. CAISO provides its rationale for scheduling any such Outages.

5. Management of Forced Outages

In the *Management of Forced Outages* section you will find the following information:

- A description of how Immediate and Imminent Forced Outages are processed in the Outage Management business processes
- A description of how a request for an extension of an Approved Maintenance Outage is processed in the Outage Management business processes

Tariff Section 9.3.10, Forced Outages

5.1 Immediate Forced Outage

Any Participating Generator or Participating TO, upon identification of a situation likely to result in a Forced Outage within the next 24 hours unless immediate corrective action is taken, where such action requires the removing from service or restricting an operating Generating Unit or removing a transmission facility from service, must communicate directly with CAISO Control Center as set forth in the Emergency procedures of the CAISO Operating Procedure (See Section 1.5, References). This situation is referred to as an Immediate Forced Outage.

An Immediate Forced Outage is directed to the CAISO Control Center.

5.2 Imminent Forced Outage

Any Operator, upon identification of a situation likely to result in a Forced Outage but of a nature not requiring a removal from service until sometime more than 24 hours in the future is subject to the provisions of Sections 4 and 5 of this BPM with respect to any necessary Outage except the timing requirements for notification in those sections which are waived and the request is expedited by CAISO provided notice is given as soon as possible. This situation is referred to as an Imminent Forced Outage.

5.3 Extended Scheduled Outage

If a Participating Generator or Participating TO wishes to continue to perform maintenance work beyond the date and time specified in an Approved Maintenance Outage, the Owner may submit a request to CAISO to extend the Approved Maintenance Outage. Following review of the request for an extension, CAISO Control Center personnel can approve an extension to an



Approved Maintenance Outage under conditions specified in the CAISO Operating Procedures (See Section 1.5, References) which are summarized below:

- CAISO is notified at least two hours before the facilities subject to the Outage are scheduled to return to service
- The extension is not expected to increase Congestion.
- No other planned Outages or derates are affected by the extension

If a request for an extension is not approved, the portion of the Outage that extends beyond the completion date and time of the Approved Maintenance Outage is considered a Forced Outage.

6. Communication of Maintenance Outage Information

In the *Communication of Maintenance Outage Information* section you will find the following information:

- A description of the need for a single point of contact for communication purposes
- A description of methods of communication to be used as a part of the Outage Management business processes
- A brief description of the SLIC system

Tariff Sections 9.3.4, Single Point of Contact and 9.3.5, Method of Communication

6.1 Single Point of Contact

All communications concerning a Maintenance Outage request or a request to confirm or change an Approved Maintenance Outage must occur between CAISO and the designated single point of contact for each Participating Generator or Participating TO. The Participating Generator or Participating TO must provide in its initial request and any subsequent changes to its Master File, the identification of the single point of contact who is responsible for all Outage Management related activities. This identification is confirmed in all communications with CAISO in relation to Outage requests, including any request to CAISO for confirmation, change or Final Approval of an Outage.

6.2 Method of Communication & Confirmation

This section includes a discussion of the primary and backup mechanisms to communicate Outage Management information, a discussion of the need for some communications to be conducted with Control Center personnel and a brief description of the SLIC application.

6.2.1 Primary Mechanism

The SLIC application is the primary method of communicating Outage Management related information. The SLIC application, which is described in more detail in Section 6.2.4 of this BPM, provides an automated mechanism for parties and CAISO to communicate the information required for all aspects of Outage Management from submittal of requests under the long-range planning process, to requesting and receiving an extension to an Approved



Maintenance Outage. The SLIC application provides both a mechanism to communicate as well as a mechanism to confirm the receipt of information from users and from CAISO.

6.2.2 Backup Mechanism

In the event that the SLIC application is not operational, emergency capabilities are used to communicate with CAISO. The emergency capabilities that can be used as a back-up if the SLIC application is unavailable include:

- Electronic format (such as E-mail or FTP file)
- Voice communication

Under emergency conditions when SLIC is inoperable, forms included in the CAISO Operating Procedure (See Section 1.5, References) are used to facilitate communication with the CAISO. Under these conditions, CAISO confirms receipt of requests and provides notice of rejection, approval, and other communications using the same format as the requester.

6.2.3 Voice Communication with Control Center Personnel

As discussed in this BPM, some Outage Management related communications by or with CAISO Control Center personnel are conducted on the telephone. These communications are described in detail in CAISO Operating Procedures (See Section 1.5, References). Examples of communications with Control Center personnel that must be conducted over the phone are:

- Requests for Final Approval of Generation Outages are made directly via phone to Control Center personnel
- Requests for Final Approval of CAISO Controlled Grid and Interconnections are made directly via phone to Control Center personnel

6.2.4 Use of SLIC

The SLIC application is a secure software system that enables parties to interact with CAISO to complete the various transactions included in the Outage Management business processes. The SLIC application includes a web client version for use by an individual and an Application Program Interface (API) version for use in computer-to-computer data transfers. Using SLIC, a Participating TO or Participating Generator or others can perform the following functions:



- Submit a request for an new Outage
- Receive confirmation of receipt of request from CAISO OCO
- Obtain status of an Outage
- Enter Outage Cause Codes (NERC GADS, reason for Outage)
- Update an Outage
- Change PMIN and Ramp Rates
- Search the database of completed, scheduled or active Outages. This function allows a generator or transmission owner to review only their data and not the data of other owners
- User instructions are available on the CAISO Website at:
 - <http://www.caiso.com/docs/2005/10/28/200510281047542112.html>

Other functions provided for in SLIC are listed in the SLIC materials shown in Section 1.5, References, of this BPM.

7. Records & Reports

In the *Records & Reports* section you will find the following information:

- A description of the Outage records retention provided for by CAISO and the access provided to those records
- A description of the various reports related to Outage Management that CAISO produces

Tariff Sections 9.3, Coordination of Outages and Maintenance and 9.5, Records

7.1 Records of Approved Maintenance Outages

CAISO OCO maintains a record of each Approved Maintenance Outage as it is implemented. Such records are available for inspection at CAISO OCO by Facility Owners of Generating Units

or sections of CAISO Controlled Grid and Interconnections, or their designated representatives. Only those records pertaining to the equipment or facilities owned by the Facility Owner are made available for inspection at CAISO OCO. Such records are only made available provided notice is given in writing to CAISO at least 15 days in advance of the requested inspection date.

7.1.1 Access to Approved Maintenance Outages

CAISO publishes approved CAISO Controlled Grid facility or Interconnection Outages on its OASIS Website 30 days prior to the Outage. A daily Transmission Outage Report showing Planned Outages for the next seven days is available on the CAISO Website. A daily snapshot of Generation Outages, both Planned and Unplanned are published on the CAISO Website each day at 1515.

7.2 System Adequacy Reports

CAISO publishes reports on an annual and a quarterly basis containing a comparison of the projected aggregate Generation capacity compared with the forecast of peak Demand. The peak Demand is calculated as described in the *BPM for Market Operations*.

CAISO publishes annual and monthly system analysis reports as described in the following sub-sections.

7.2.1 Rolling Twelve Month System Adequacy Forecast

On a quarterly basis, and approximately eight weeks after receiving the annual or updated long-range Outage requests from Participating Generators and Participating Transmission Owners, CAISO publishes on the CAISO Website a forecast comparing the aggregated weekly peak Generation and interconnection capacity to the weekly peak forecast Demand for the next 52 weeks.

7.2.2 Monthly System Adequacy Forecast

On a monthly basis, and approximately one week prior to the start of each month, CAISO publishes a forecast on CAISO Website that compares the aggregated daily peak Generation and interconnection capacity to the weekly peak forecast Demand for the next month. The weekly peak Generation capacity used in the comparison reflects all approved Outages to date.

7.3 Reporting of Forced Outages

Tariff Section 9.3.10.4, of Section 9.3.10 Forced Outages

Facility Owners or their designated representative must submit a report of any Forced Outage or reduction of capacity of a Generating Unit.

Changes in availability of 10 MW or 5% of Pmax (whichever is greater) lasting 15 minutes or longer must be reported to the ISO. These reports are due to the ISO within 60 minutes of discovery, and are required only to include effective time and MW availability.

Additionally, a more detailed Forced Outage Report is required for derates of 40 MW or 10% of Pmax (whichever is greater) lasting 15 minutes or longer. These reports are due to the ISO within two business days after the availability report has been submitted.

Either the SLIC Web-Client or the SLIC Real-Time Availability Monitoring (which utilizes the SLIC API) may be used to report availability changes. The SLIC is also used to later submit any required Forced Outage Report.

7.3.1 Resource Adequacy Resources with Pmax between 1.0 MW and 10 MW

CAISO Tariff Section 40.9.5 requires Resource Adequacy Resources with a Pmax between 1.0 MW and 10 MW that are not required to report Forced Outages within 60 minutes of discovery

to provide equivalent availability-related information at the end of the calendar month. Such resources must use the SLIC Web-Client to report all Forced Outages or de-rates and temperature-related ambient de-rates.

Resource Adequacy Resources with a Pmax between 1.0 and 10 MW are not required to report their availability near real-time, but must provide Forced Outages or de-rates and temperature-related ambient de-rates no later than 3 business days after the end of the calendar month. Operating Procedure T-113, Scheduled and Forced Outages, section 13.1 Immediate Forced Outages or Derates, RA Resources provides details on how Resource Adequacy Resources with a Pmax between 1.0 MW and 10 MW should report availability. These Resource Adequacy Resources are subject to Non-Availability Charges and Availability Incentive Payments as described in Section 40.9.6 of the CAISO Tariff and Section 8 of the BPM for Reliability Requirements.

7.3.2 Proxy Demand Resource Outage Reporting

Proxy Demand Resource (PDR) outages and derates shall be managed and reported to SLIC in the same manner as generator resources. Multiple outage types will be supported, including, but not limited to, PMax Derates and Ramp Rate Derates.

A Proxy Demand Resource will submit a PMax derate to indicate that a PDR is unavailable. If a PDR submits a PMax derate, the day will be considered an outage event day for the purpose of

the baseline calculation. The only PMax derate permitted for PDR is a derate to 0 MW's. (i.e. a PDR is either 100% available or 0% available, there are no partial derates of PDRs)

A PDR will submit a Ramp Rate derate to indicate that a PDR is unavailable to fulfill a previously awarded A/S award.

7.3.3 Reliability Demand Response Resource

Reliability Demand Response Resource (RDRR) outages and derates shall be managed and reported to SLIC in the same manner as generator resources. Multiple outage types will be supported, including, but not limited to PMax Derates and Ramp Rate Derates.

A Reliability Demand Response Resource will submit a PMax derate to indicate that a RDRR is unavailable. If a RDRR submits a PMax derate, the day will be considered an outage event day for the purpose of the baseline calculation. Full and partial outage derates along with Transmission and Distribution system Emergency Dispatch Outages may be declared by the operating Scheduling Coordinator.

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7.3.4 Non Resource Specific RA Resources. Non Resource Specific RA outages and derates shall be managed and reported to SLIC in the same manner as generator resources, per ISO tariff section 9, and ISO Procedure 3210 section 3.2 Generation Facility Outages. Transmission interchange schedule changes shall be reported to the ISO real-time scheduling desk and balanced as required in CAISO procedures 2410, 2510, 2520 and 1610.

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7.4 Questionable Outage Report to FERC

Tariff Section 9.3.10.5, of Section 9.3.10 Forced Outages

CAISO is required to report to FERC the occurrence of any questionable Forced Outage behavior. The factors that CAISO considers in determining whether an Outage is questionable are detailed in the Tariff Section 9.3.10.5 and summarized below:

- If the Forced Outage coincide with certain market conditions that such that prices or cost of payments may be influenced
- If the Forced Outage coincided with a change of bids into CAISO Markets
- If CAISO recently rejected a request for an outage for the Generating Unit experiencing the Forced Outage
- If the timing or content of the notice is inconsistent with subsequent reports
- If the Forced Outage or its duration is inconsistent with history or past performance
- If the Forced Outage creates or exacerbates Congestion or is extended with little or no notice
-
- If the Operator has other alternatives or did not take reasonable action to minimize the Outage
- If the Operator fails to provide an explanation to CAISO within the required 48 hours or failed to provide sufficient information or access to the Generating Unit.



Appendix A.

ISO NON-DISCLOSURE AND USE OF INFORMATION AGREEMENT

FOR SHARING OF GENERATION AND ELECTRIC TRANSMISSION OUTAGE INFORMATION WITH A NATURAL GAS UTILITY

This Non-Disclosure and Use of Information Agreement for Sharing of Generation and Electric Transmission Outage Information with a Natural Gas Utility ("Agreement") is made and entered into as of this _____ day of _____, 20____, by and between the California Independent System Operator and _____.



Operator Corporation ("ISO") and _____ ("Receiving

Party") (individually the ISO and Receiving Party are each a "Party" and collectively "Parties").

WHEREAS, pursuant to ISO tariff Section 20.4(c)(iv), the Receiving Party seeks to obtain certain confidential or proprietary information from the ISO (i) regarding Maintenance Outages or Forced Outages of natural gas-fired generation resources or of elements of the ISO-Controlled Grid, or (ii) necessary for day-to-day coordination and longer-term planning of gas transmission and pipeline outages (collectively, "Outage Information"); and

WHEREAS, pursuant to ISO tariff Section 20.4(c)(iv), the ISO is willing to provide such Outage Information to the Receiving Party, under suitable contractual limits and protections concerning the disclosure and use of confidential or proprietary information,

NOW, THEREFORE, in consideration of the mutual covenants in this Agreement, the ISO and the Receiving Party agree as follows:

1. Purpose, Scope and Definition. The purpose of this Agreement is to permit the ISO to share Outage Information that constitutes Confidential Information with the Receiving Party, in order to maintain reliable operation of the ISO Balancing Authority Area as set forth in ISO tariff Section 20.4(c)(iv). Confidential Information under this Agreement consists of (i) all written materials, including electronic materials, marked "Confidential," "Proprietary," or with words of similar import provided to the Receiving Party pursuant to this Agreement, (ii) all observations or equipment (including computer screens) and oral disclosures related to the ISO's systems, operations, and activities that are indicated as such at the time of observation or disclosure, respectively, and (iii) portions of documents, records, and other material forms or representations, including any electronic materials, which the Receiving Party may create, including but not limited to handwritten notes or summaries, that contain or are derived from the Confidential Information described in sections (i) and (ii) of this paragraph.

2. Non-Disclosure. Subject to Paragraph 4 below, the Receiving Party shall keep Confidential Information in strict confidence and shall not disclose such information or otherwise make it available, in any form or manner, to any other person or entity, including but not limited to, generating or marketing affiliates (a "third party") other than its employees or consultants, without the prior written consent of the ISO. Notwithstanding any other provision of this Agreement, disclosure of Confidential Information by the Receiving Party, including to its employees and consultants, shall be subject to and limited by FERC's Standards of Conduct requirements. The Receiving Party will cause each of its employees and consultants who will have access to Confidential Information to acknowledge that they have read this Agreement and agree to abide by all of its terms regarding use and disclosure of Confidential Information by execution of Exhibit A. It is the ongoing responsibility of the Receiving Party to ensure that (i) each Exhibit A is accurate, (ii) each Exhibit A permits access only to a current employee or consultant of the Receiving Party, and (iii) each new Exhibit A and any notice of cancellation of an Exhibit A is immediately submitted to the ISO. The Receiving Party shall immediately report to the ISO any breach of this Agreement.

3. Use of Confidential Information.

(a) The Receiving Party shall use Confidential Information received hereunder only for the purposes identified herein, including using Confidential Information, subject to the restrictions on disclosure, to manage, coordinate, plan, forecast, and/or 11



schedule outages, maintenance, repairs, and/or curtailment of its gas

transmission pipeline and storage system.

(b) It is understood and agreed to by the Receiving Party that Confidential Information constitutes confidential and/or proprietary information of the ISO. The Receiving Party shall maintain Confidential Information in strict confidence and shall not disclose, duplicate, or otherwise reproduce Confidential Information (other than for back-up purposes), directly or indirectly, in whole or part. Further, the Receiving Party understands that California Public Utilities Code Section 352.7 may apply to the terms of this Agreement. Nothing in this Agreement, however, shall be deemed to constitute an agreement, admission, or waiver that California Public Utilities Code Section 352.7 applies.

4. Exceptions to Non-Disclosure. Notwithstanding Paragraph 2 above, a party to this Agreement shall not have breached any obligation under this Agreement if Confidential Information is disclosed to a third party when Confidential Information:

(a) was in the public domain at the time of such disclosure or is subsequently made available to the public consistent with the terms of this Agreement; or

(b) had been received by the Receiving Party at the time of disclosure through other means without restriction on its use, or had been independently developed by the Receiving Party as shown through documentation; or

(c) is subsequently disclosed to the Receiving Party by a third party without restriction on use and without breach of any agreement or legal duty; or

(d) subject to the provisions of Paragraph 5, is used or disclosed pursuant to statutory duty or an order, subpoena or other lawful process issued by a court or other governmental authority of competent jurisdiction.

5. Notice of Pending Third-Party Disclosure.

(a) In the event that a court or other governmental authority of competent jurisdiction issues an order, subpoena, or other lawful process requiring the disclosure of Confidential Information, the Receiving Party shall notify the ISO immediately upon receipt thereof to facilitate the ISO's efforts to prevent such disclosure, or otherwise preserve the confidentiality of the Confidential Information.

(b) The Receiving Party shall not be in violation of this Agreement if it complies with an order of a court or governmental authority, or a public records law or regulation, requiring disclosure of the Confidential Information, after the ISO either has unsuccessfully sought to maintain the confidentiality of such information as provided herein, has notified the Receiving Party in writing that it will take no action to maintain such confidentiality, or has failed to act within a reasonable period of time.

6. Term. This Agreement shall remain in effect unless and until either Party provides ten (10) days prior written notice to the other Party. Termination shall not extinguish any claim, liability, or cause of action under this Agreement existing at the time of termination.

7. Provisions Surviving Termination. The provisions of Paragraphs 2, 3, 4, and 5 shall survive the termination of this Agreement for a period of ten (10) years. The provisions of 12



Paragraph 8 shall continue after termination until satisfied.

8. Return or Destruction of Confidential Information. Upon termination of this Agreement, all Confidential Information in the possession or control of the Receiving Party, including its employees and consultants, shall be returned to the ISO, including all copies of such information in any form whatsoever, unless otherwise instructed in writing by the ISO. However, if the Confidential Information is retained in the computer backup system of the Receiving Party, the Confidential Information will be destroyed in accordance with the regular ongoing records retention process of the Receiving Party, or the Confidential Information will be retained under applicable law, court order, stock market regulation, or competent authority of any governmental body. In lieu of return, the Receiving Party may certify to the ISO in writing that all such information, in any form whatsoever, has been destroyed. Notwithstanding the foregoing, non-destruction of any operating or outage plans of the Receiving Party that contain, incorporate, or reflect Confidential Information shall not be deemed to violate this Agreement, provided that the Confidential Information contained therein is not disclosed or used in violation of the other terms of this Agreement.

9. Notices.

(a) Representatives and Addresses. All notices, requests, demands, and other communications required or permitted under this Agreement shall be in writing, unless otherwise agreed to by the parties, and shall be delivered in person or sent by certified mail, postage prepaid, by overnight delivery, by electronic mail, or electronic facsimile transmission. In the case of electronic mail or electronic facsimile transmission, an original shall be sent immediately thereafter by postage prepaid mail. All communications shall be properly addressed as follows:

Receiving Party:

Name of Entity: _____

Name of Representative (person/position): _____

Address: _____

Telephone: _____

Facsimile: _____

Email: _____

ISO:

(person/position) _____

California Independent System Operator Corporation

250 Outcropping Way

Folsom, CA 95630

Telephone: (916) - _____

Facsimile: (916) - _____

E-mail: _____

With a copy to:

Legal Department

California Independent System Operator Corporation

250 Outcropping Way

Folsom, CA 95630

Telephone: (916) 608-7021

Facsimile: (916) 608-7296

(b) Changed Representatives and Addresses. A party hereto may from time to time change its representative or address for 13



the purpose of notices to that party by a similar notice specifying a new representative or address, but no such change shall be deemed to have been made until such notice is actually received by the party being so notified.

(c) Effective Date of Notices. All notices and other communications required or permitted under this Agreement that are addressed as provided in this Paragraph 9 shall be effective upon delivery if delivered personally, by overnight delivery, by electronic mail, or by facsimile transmission; if delivered by overnight delivery, such notices shall be effective three (3) days following deposit in the United States mail, postage prepaid.

10. Complete Agreement; No Other Rights.

(a) This Agreement contains the complete and exclusive agreement of the parties with respect to the subject matter thereof, and supersedes all discussions, negotiations, representations, warranties, commitments, offers, contracts, and writings prior to the date of this Agreement, with respect to its subject matter. No change to this Agreement shall be effective unless agreed to in writing by the parties hereto. Any conflict between the language of this Agreement and any mark, stamp, annotation, or other language identifying something received hereunder as Confidential Information shall be resolved in favor of this Agreement.

(b) This Agreement is not intended to create any right in or obligation of any party or third party other than those expressly stated herein.

(c) Nothing in this Agreement sets a precedent, nor creates any right or obligation, with respect to any future access, disclosure, or production of Confidential Information.

11. No Warranties or Representations. Any Confidential Information disclosed by the ISO under this Agreement carries no warranty or representation of any kind, either express or implied. The Receiving Party shall not be entitled to rely on the accuracy, completeness, or quality of Confidential Information, even for the purpose stated in Paragraph 1.

12. Injunctive Relief. The Receiving Party agrees that, in addition to whatever other remedies may be available to the ISO under applicable law, the ISO shall be entitled to obtain injunctive relief with respect to any actual or threatened violation of this Agreement by the Receiving Party or any third party. The Receiving Party agrees that it shall bear all costs and expenses, including reasonable attorneys' fees, that may be incurred by the ISO in enforcing the provisions of this paragraph, only if the ISO prevails in the litigation.

14. Governing Law. This Agreement is made in the State of California and shall be governed by and interpreted in accordance with its laws.

15. Assignment. This Agreement shall be binding upon the parties, their successors, and assigns. The Receiving Party shall not assign this Agreement without the ISO's prior written consent.

16. Construction of Agreement. Ambiguities or uncertainties in the wording of this Agreement shall not be construed for or against any party, but shall be construed in the manner that most accurately reflects the parties' intent as of the date they executed this Agreement.



17. Signature Authority. Each person signing below warrants that he or she has been duly authorized by the party for whom he or she signs to execute this Agreement on behalf of that party. 14 18. Counterparts. This Agreement may be executed in two or more counterparts, all of which shall be considered one and the same Agreement.
19. Consistency with Federal Laws and Regulations. Nothing in this Agreement shall compel any person or federal entity to (1) violate federal statutes or regulations, or (2) in the case of a federal agency, to exceed its statutory authority, as defined by any applicable federal statutes, regulations, or orders lawfully promulgated thereunder. If any provision of this Agreement is inconsistent with any obligation imposed on any person or federal entity by federal law or regulation to that extent, it shall be inapplicable to that person or federal entity. No person or federal entity shall incur any liability by failing to comply with a provision of this Agreement that is inapplicable to it by reason of being inconsistent with any federal statutes, regulations, or orders lawfully promulgated thereunder; provided, however, that such person or federal entity shall use its best efforts to comply with the Agreement to the extent that applicable federal laws, regulations, and orders lawfully promulgated thereunder permit it to do so.
IN WITNESS WHEREOF, the parties have executed this Agreement as of the date set forth above.
CALIFORNIA INDEPENDENT SYSTEM OPERATOR CORPORATION

By: _____
Name: _____
Title: _____
Date: _____
RECEIVING PARTY:

[Company Name]
By: _____
Name: _____
Title: _____
Date: _____

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